



A Comparative Machine Learning Analysis of Geospatial Vulnerability in UNESCO World Heritage Sites

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Introduction & Motivation



The Goal

Preserving UNESCO World Heritage Sites of "Outstanding Universal Value" for future generations.



The Threat

Sites are increasingly threatened by armed conflict, climate change, rapid urbanization, and socio-economic displacement.



The Problem

Current "In Danger" designations are highly reactive, strictly occurring *after* structural degradation has already happened.



The Solution

Transitioning to proactive preservation using predictive machine learning frameworks integrated with geospatial data.

The Data Challenge: The Accuracy Paradox

Global Inventory

Analysis covers over 1,100 officially listed UNESCO sites globally.

~96%

Classified "Safe"

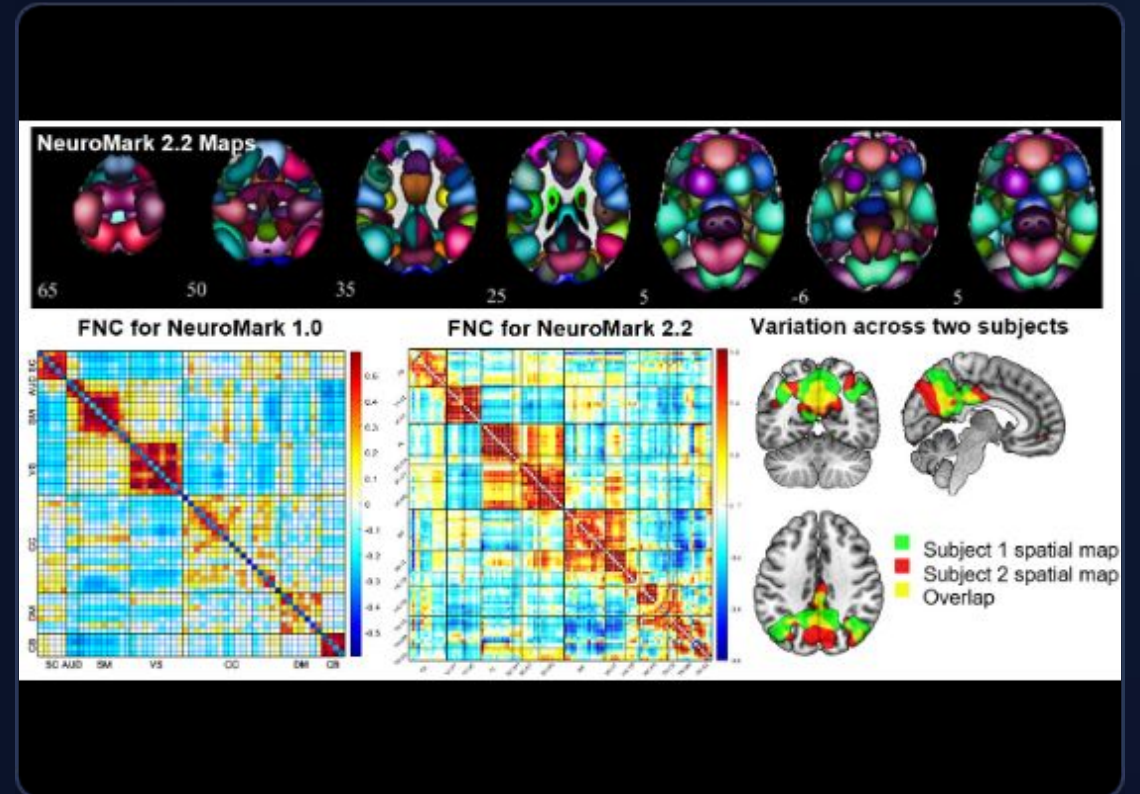
~4%

Classified "In Danger"

The Accuracy Paradox

- Standard ML algorithms optimize for global accuracy.
- A trivial model guessing "Safe" every time hits **96% accuracy** but provides zero predictive utility for conservationists.

Objective: Develop a model that penalizes minority misclassification without generating artificial coordinate artifacts.



Methodology & Models

📦 Features Used

Continuous spatial metrics (Latitude/Longitude), temporal variables (Inscription Year), and categorical attributes (Cultural, Natural, Mixed).

📊 Models Evaluated

📈 **Logistic Regression:** A cost-sensitive linear baseline.

🏗️ **Gradient Boosting (GBDT):** A sequential ensemble tree (high global optimization).

🌲 **Tuned Random Forest:** A parallel ensemble utilizing bootstrap aggregating with algorithmic-level class weighting.



Results: The Baseline & The Paradox

Logistic Regression (Baseline)

66.4
ACCURACY
%

57.9
RECALL
%

Takeaway: High sensitivity to danger, but suffers from a massive false-positive rate (precision: 8.1%).

Gradient Boosting (The Paradox)

91.6
ACCURACY
%

0.0
RECALL
%

Takeaway: Perfect illustration of the accuracy paradox. The model minimized global loss by ignoring the endangered sites entirely.

Results: The Optimal Solution

Tuned Cost-Sensitive Random Forest

By utilizing balanced class weighting during node splits, the model successfully balanced the trade-off between precision and recall.

Model Architecture	Global Accuracy	Minority Recall	Macro F1-Score
Logistic Regression (Baseline)	66.4%	57.9%	0.14
Gradient Boosting (GBDT)	91.6% (Highest)	0.0% (Failed)	0.48

Table 1: Model Comparison Results illustrating the trade-off between Accuracy and Recall.

Feature Importance & Geospatial Reality

What Drives Endangerment?

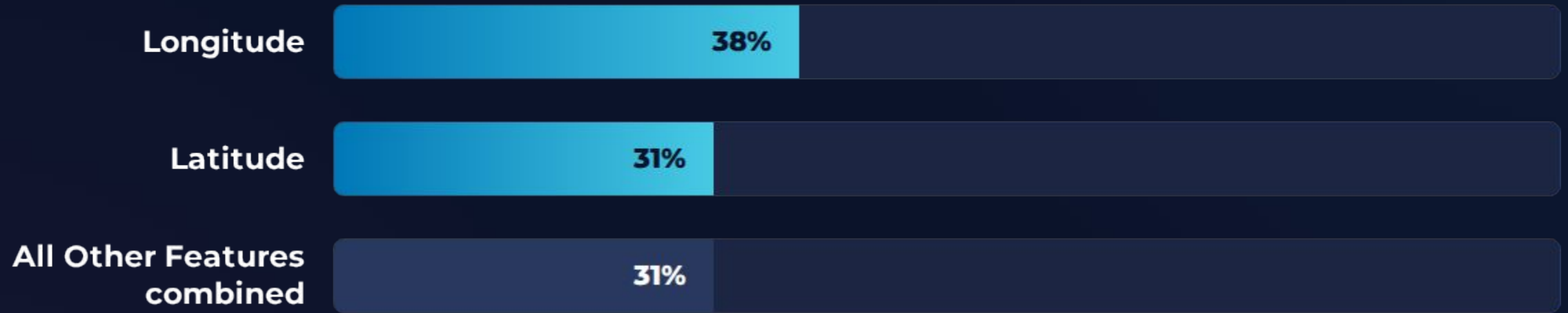


Figure 1: Feature Importance Chart from Tuned Random Forest

The Insight

Spatial coordinates account for **69% of total predictive importance**. This mathematically validates that a site's absolute location is the strongest proxy for regional instability, climate exposure, and geopolitical conflict.

Conclusion & Future Integration

Summary: ML can predict heritage risk, but researchers must explicitly design around severe class imbalance.

Architecture: Cost-sensitive Random Forests provide operational utility over standard boosting methods.

Future Policy Application

1

ML Model Prediction

Algorithm flags high-risk spatial coordinates based on continuous data tracking.

2

Satellite Verification

Conservationists task high-resolution Earth Observation (EO) satellites to verify structural integrity at those



Questions & Discussion

Thank you for your time and attention.

Special thanks to Advisor Prof. Sibghat Ullah Bazai.

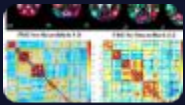
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